

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 2ND SEMESTER SECOND INTERNAL EXAMINATIONS,
JULY 2024
23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++(IT-A/IT-B /ECE-B)
(2023 Scheme)

TIME: 2 hrs**MAX. MARKS: 30****Course Outcomes**

On completion of this course the student will be able to:

CO1	Understand the basic concepts of Object-Oriented Programming.
CO2	Describe the object-oriented paradigm with concepts of streams, classes, functions, data and objects.
CO3	Implement object-oriented programming constructs like encapsulation, inheritance and polymorphism.
CO4	Understand dynamic memory management techniques using pointers, constructors, destructors, etc.
CO5	Identify classes including data, methods and the relationship among the classes from a given problem statement and solve the problem using object oriented constructs in C++.

PART A**(Answer all questions)****(5 x 2= 10 Marks)**

Q.No	Questions	Marks	BL	CO	PO
I (a)	Discuss the role of access specifiers in inheritance and show their visibility when they are inherited as public, private and protected.	2	L2	3	1,2,3,12
(b)	What is a virtual base class?	2	L1	3	1,2,3,12
(c)	Briefly explain different methods to open a file.	2	L2	4	1,12
(d)	Explain different file modes in C++.	2	L2	4	1,12
(e)	Write a C++ program using function template to find the product of two integer and floating point types of data.	2	L3	4	1,12

PART B**(Answer the following questions
choosing either the first OR second option)****(1 x 10= 10 Marks)**

Q.No.	Questions	Marks	BL	CO	PO
II (a)	What is inheritance? Explain different types of inheritance.	3	L2	3	1,2,3,12
(b)	Write a C++ program to calculate the total marks of each student of a class in Physics, Chemistry, Mathematics and average marks of the class. The number of students in the class are entered by the user. Create a class named marks with data	7	L3	5	1,2,3,5,6, 9,12

	members for, roll number, name and marks. Create three other classes inheriting the marks class, namely Physics, Chemistry and Mathematics which are used to define marks in individual subjects of each subject. Which type of inheritance does this model belong to?				
OR					
III (a)	Briefly explain operator overloading? Write a C++ program to overload - (unary minus operator: it represents negative numbers, if a variable x has the value 7, then -x would have the value -7) using the overloading function.	6	L2	3	1,2,3,12
(b)	Explain pure virtual functions in C++.	4	L2	3	1,2,3,12
(1 x 10 = 10 Marks)					
IV (a)	What is a file pointer? Explain the functions related to file pointers.	5	L2	4	1,12
(b)	Write a C++ program containing a possible exception. Use a try block to throw it and a catch block to handle it properly.	5	L3	2	1,2,12
OR					
V (a)	Discuss in detail about exception handling in C++.	5	L2	2	1,2,12
(b)	Write a C++ program to read any data from the user, write it to a file and display it.	5	L3	4	1,12

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
4	60	36	-	-	-